

MoodSyncAI

Multi-Modal Sentiment & Emotion Analyser

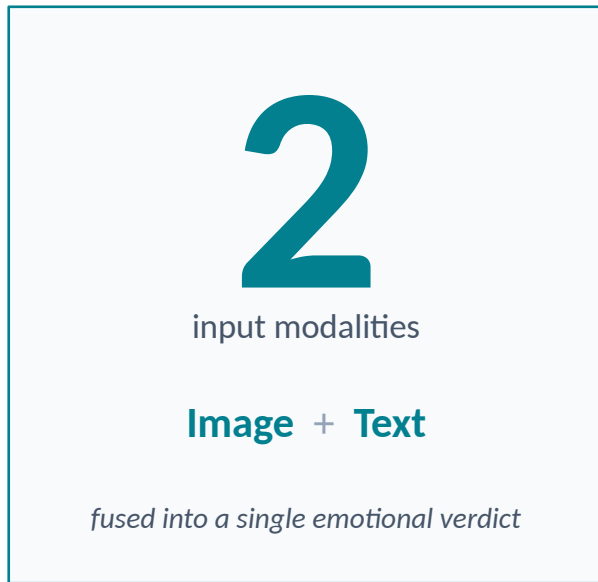
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Data Analytics 3 — Final Project

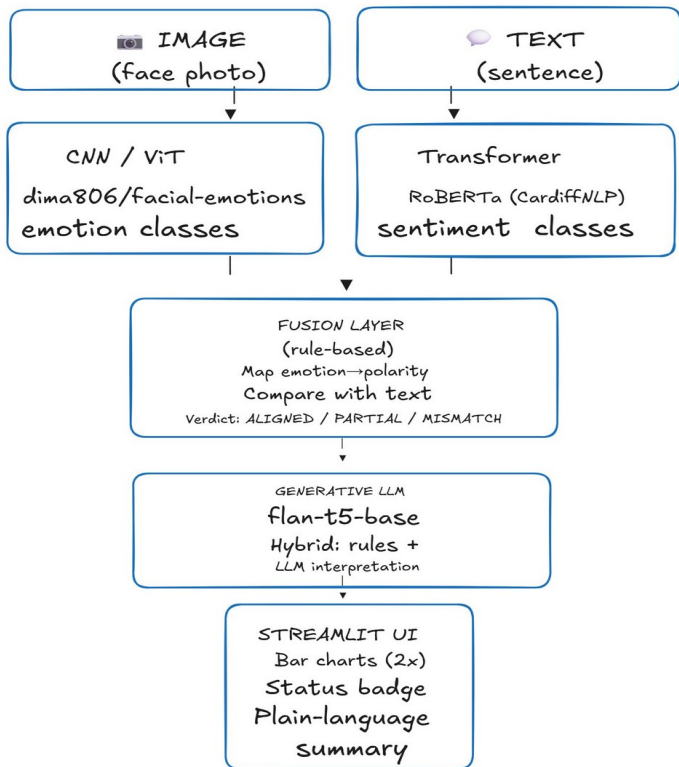
SRH Hamburg · Prof. Dr. Gayan de Silva · May 2026

The Problem: Words Don't Always Match Faces

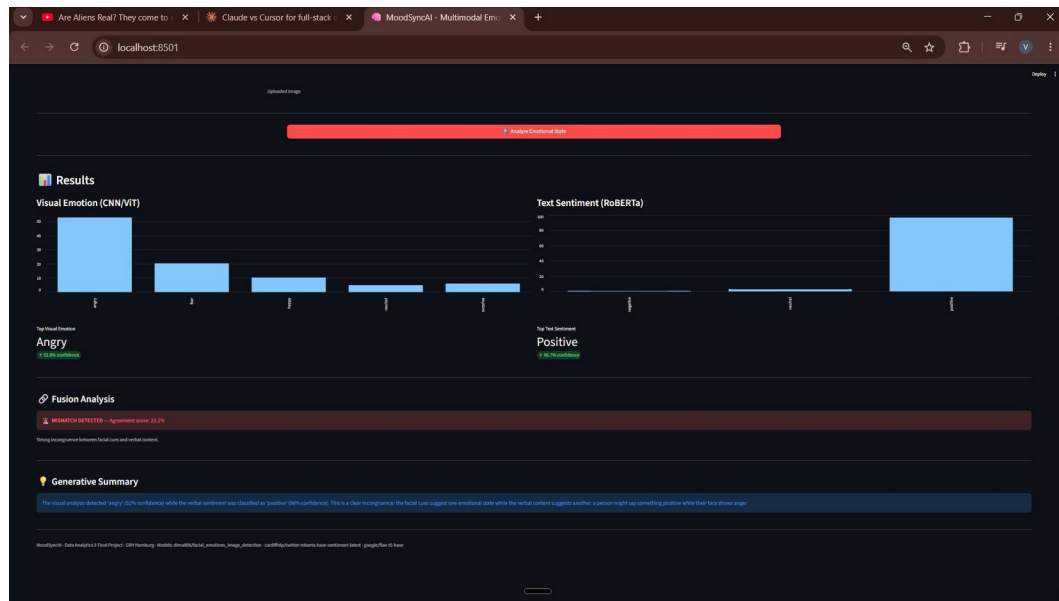
- A person may say "I'm fine" while their face shows distress.
- Single-modality emotion analysis misses these incongruences.
- **Goal: combine visual and verbal signals to detect mismatch and produce a plain-language explanation.**



System Architecture



Demo: Detecting an Emotional Mismatch



Streamlit dashboard, MISMATCH DETECTED state

Generative summary explains the incongruence in plain language.

INPUT

Image: person showing anger

Text: "No, I think the project is going really well."

OUTPUT

Visual emotion angry (52.8%)

Text sentiment positive (96.7%)

Fusion status **MISMATCH**

Agreement 25.2%

Extended Features Implemented

Three of the optional features from section 4 of the assignment brief.

EXT. FEATURE 1	EXT. FEATURE 2	EXT. FEATURE 3
Webcam Input Streamlit st.camera_input captures a live still from the user's webcam. No separate upload step required. Pipes directly into the existing CNN pipeline. Coexists with file upload via a radio toggle.	Audio + Whisper Streamlit st.audio_input records speech in-browser. Whisper-tiny (~39 MB) transcribes the clip; the transcript auto-fills the text box. Feeds the existing RoBERTa sentiment model.	ViT Attention Rollout Visualises which face regions the ViT attended to (Abnar & Zuidema, 2020). 12-layer attention rollout with residual identity; CLS-to-patch attention reshaped to 14x14 and overlaid as a JET heatmap. Chosen over Grad-CAM (CNN-only) for this transformer architecture.

Challenges I Overcame & What's Next

CHALLENGES OVERCOME

1. Generative summary hallucination

flan-t5-base parroted prompts when asked for full 3-sentence summaries. Solved with a hybrid: rule-based factual sentences + LLM for interpretation only.

2. Multimodal alignment without labels

No paired image-text dataset with mismatch labels. Used a deterministic emotion-to-polarity mapping and rule-based fusion instead of training a fusion network.

3. ViT attention without exposed weights

HuggingFace pipeline hides attention tensors. Re-loaded the model via AutoModelForImageClassification with output_attentions=True to extract them.

WHAT'S NEXT

- **Real-time video timeline**

Show emotion changes frame-by-frame as the user speaks.

- **Learned fusion network**

Replace rules with a small neural net once paired labelled data is available.

- **Cloud deployment**

Host on Hugging Face Spaces with a public URL (requires paid tier for 3 GB model footprint).

- **Domain-specific fine-tuning**

Re-train RoBERTa on workplace dialogue rather than tweets.